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Trajectories reconstruction of spinning baseball pitches by three-point-based algorithm

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ABSTRACT

In this work, a method to reconstruct trajectories by obtaining the initial conditions, velocity and angular velocity, from spinning throws is designed. It is based on considering Magnus effect can be separated from rest of forces that define the dynamics of the ball, such assumption is supported by an energetic analysis. Thus, the methodology consists in solving the two-point boundary value problem (BVP) of the movement equations without Magnus force and then adding its effect.

We found that only three points (ball positions as functions of time) of a trajectory are necessary to characterize it, and consequently to solve the three-point BVP. This result is applied in an algorithm based on the shooting method, which obtains the initial conditions of synthetic trajectories by minimizing the distance between points of the original trajectory and those of the proposed trajectory, with high accuracy in a short time.

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1. Introduction

1.1. Background and summary

In the context of baseball, the reconstruction of trajectories is generally applied to replay of pitches on broadcast of games. Existing methods, focus only on tracking the ball by means of tools processing all image available. They consist of three basic steps: First, an overlap of a lot of photos of the same pitch is carried out to obtain the position of the ball at different times; also it is used to transform the 3D reality into a 2D image. Then, several trajectories are proposed using probabilistic methods, database comparisons, and/or color and region filtering based on pixel analyses [1–5]. Thus, some discrete trajectories are filled with the same methods or by a parameter estimation [6]. Lastly, chosen trajectories are generally compared with mathematical models of the ball dynamics.

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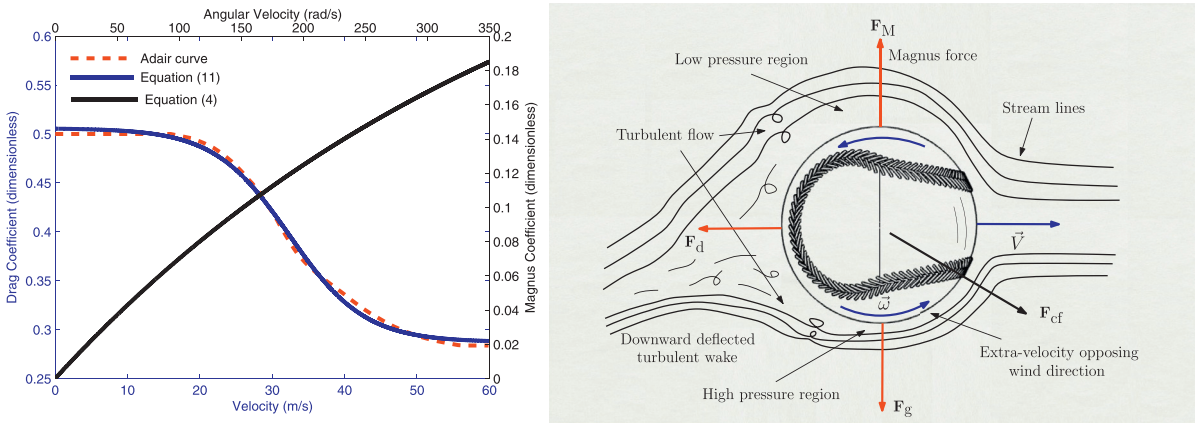

Fig. 1. Left: Drag (C_d) and Magnus (C_M) coefficients. Red dashed line represents the Adair estimation (data were extracted using tools of Matlab R2013a program) for C_d [15]. Blue solid line approximates Adair curve (Eq. (11)), while black line fits to Briggs experimental data [29] using Eq. (4). Right: Diagram of forces and stream lines of a rotating baseball moving to the right with an angular velocity $\boldsymbol{\omega}$. The forces involved are gravitational ($\mathbf{F}_g$), drag ($\mathbf{F}_d$) and Magnus ($\mathbf{F}_M$). Centrifugal force ($\mathbf{F}_{cf}$) is also included in. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In this way, such methods usually do not deal with the movement equations of the ball so that a large number of ball positions data is required only to complete the trajectory but initial conditions of the ball are not obtained. However, they have been reported feasible and bringing good results for tracking the ball. On the other hand, the methodologies involving to solve the BVP of the ball movement equations are poorly viable for their use because of the data or the equipment required [7,8].

We designed a method to reconstruct trajectories by obtaining the initial conditions of spinning throws taking into account the movement equations of the ball mass center. The work is focused on pitches with initial conditions, velocity $\mathbf{V} \equiv (V_x, V_y, V_z)$ and angular velocity $\boldsymbol{\omega} \equiv (\omega_x, \omega_y, \omega_z)$, around the range of a throw made by a professional pitcher, namely, $\mathbf{V} \in [(-3, 30, -3), (3, 50, 3)]$ m/s, $|\boldsymbol{\omega}| \in [100, 310]$ rad/s [9], fixing y-axis in mound-home direction, z-axis perpendicular to the Earth's surface, and x-axis orthogonal to both y and z axes, according to the right hand rule.

This paper is structured as follows: Section 1.2 has the purpose of presenting the equations of motion of the ball. Basis of the method is to consider Magnus force can be separated from rest of forces that define the dynamics of the ball; we prove such assumption in Section 2.1 by an energetic analysis, and estimate the Magnus effect in Section 2.2 numerically. In Section 3, we describe the method (Section 3.1) and present an algorithm (Section 3.2). Results and conclusions are shown in Sections 4 and 5, respectively.

1.2. Dynamics of spinning pitches

Dynamics of a rotating baseball with mass m and moving at a velocity $\mathbf{V}$ is defined by the Newton's second law [10,11], such that

$$m\dot{\mathbf{V}} = \mathbf{F}_B + \oint_S \sigma \, dS \cdot \hat{\mathbf{n}}, \tag{1}$$

where the first term on the right side is the sum of forces acting through the volume of the ball (gravitational and Coriolis forces, and pseudo forces); whereas the second term is the net force, where its components are defined by the stress tensor σ, acting across the surface S of the ball, such it covers all forces produced as a consequence of the interaction air-ball: drag, Magnus, torque, cross and constraint forces [11].

When a ball is thrown with high spins, like those inside the range mentioned in Section 1.1, the forces caused by the asymmetries of the ball seams, cross forces, vary from positive to negative quickly and have sufficiently low magnitude to be omitted [12,13]. In addition, low decelerations on angular frequency ω – around 0.08% for curveballs [14] and 20% for golf balls after a flight of 5 s – together with the low precession of the spin axis $\hat{\boldsymbol{\omega}}$ let not consider torque forces. Moreover, constraint forces are commonly omitted because the stiffness of the ball is practically not affected when it travels through the air [15].

Thus, main forces involved in the integral of the right side of the Eq. (1) are drag and Magnus. Drag can be explained by the difference of pressure between front and back sides of the ball [16]. As shown in Fig. 1, stream lines on the back side of the ball are farther than on the front. This creates a low pressure region on the back side that causes a momentum with reverse direction to its motion. Experimental data suggest that difference of pressure and drag force are connected by an approximation proportional to the square of the ball speed [15,17,18], so that

$$F_d = \frac{1}{2}\rho A C_d V^2 \tag{2}$$

Table 1
Estimation of acceleration (α) and deflection ($d\xi$) of a baseball caused by forces acting on a professional spinning throw.

Force	Approximation	α (cm/s^2)	dξ (cm)
Gravitational	$m\mathbf{g}$	978–980	$\sim$ 122
Centrifugal (rad)	$m\Omega^2 R \sin^2\psi\ \hat{\mathbf{g}}$	0–3.4	0–0.42
Centrifugal (tan)	$m\Omega^2 R \sin\psi \cos\psi\ \hat{\mathbf{s}}$	0–1.7	0–0.21
Coriolis	$2m\mathbf{V}\times\mathbf{\Omega}$	0.4–0.7	0.05–0.09
Drag	$kC_\mathrm{d}V^2\hat{\mathbf{V}}$	650–1350	80–170
Magnus	$kC_\mathrm{M}\sin\phi V^2\hat{\mathbf{u}}$	140–850	18–105

where factor $\frac{1}{2}\rho V^2$ is the difference of pressure between front and back sides (with ρ the air density) that exists considering the ball has a transverse section front area A [16]. In turn, C_d is a dimensionless coefficient that is introduced because interaction between the front area and the stream lines go down when the speed of the ball V increases, i.e., the effective front area decreases, which makes the ball more aerodynamic. Values of such coefficient are estimated by Adair [15] considering the average of experimental data for all types of pitches and measures of baseballs in free fall; see the left-hand side of Fig. 1 by the red sigmoidal curve. Despite there exist modern approaches for the drag coefficient (Alaways [21] and Cross [22]), we have chosen the Adair approximation with the aim of having a continuous set of values for it as functions of the instantaneous speed of the ball without requiring to know the initial speed of the ball. This for the purpose of the present research. Moreover, Adair model is not far from the values mentioned in both Alaways and Cross, and in general, such model is more accepted than other models for describing the behavior of C_d [20,23,24].

Model (2) is supported by Reynolds numbers (Re) in the order of $6.5\times10^4 - 1.8\times10^5$ produced by the sizes and velocities of the ball [9,19–21]. Obtained values of Re correspond to the laminar-turbulent transition flow of air [18]. This allows to let only the quadratic approximation of a Taylor series [10] remain like in Eq. (2), and to compute

$$\mathbf{F}_\mathrm{d} = -\frac{1}{2}\rho A C_\mathrm{d} V^2 \hat{\mathbf{V}} \tag{3}$$

by definition of the drag force. On the other hand, Magnus force is present in any rotating ball (like soccer [25], cricket [18,26] and tennis [27]). Magnus effect is caused when a rough ball flies spinning through the air: a side of it opposes the air force causing a bigger resistance than in the opposite side, i.e, air velocity on the upper side is lower than at bottom side, according to right-hand side of Fig. 1. Then, since low velocities are associated with high pressure regions and vice versa [28], a low pressure region is originated at the top of the ball leading to the common Magnus effect[1] [15,18].

Similarly to drag force, experimental measures indicate that Magnus force acting in upward direction can be written as $F_\mathrm{M} = \frac{1}{2}\rho A C_\mathrm{M} V^2$ with the adaptive dimensionless coefficient of area C_M as a function of ω and V [15,20,29]. Even more, experimental studies of Nathan [20] show that C_M (the Magnus coefficient) depends mainly on the angular velocity, for ω and $\mathbf{V}$ values inside the range mentioned in Section 1.1. This permits to complete the model expressing the Magnus coefficient by the exponential relation

$$C_\mathrm{M}(\omega) = 3.19\times10^{-1}\left[1 - e^{-2.48\times10^{-3}\omega}\right], \tag{4}$$

which originally was employed to fit C_M values for golf balls, but that is also close to experimental data reported by Briggs [29], as mentioned in Robinson and Robinson's works [18,30]. Expression (4) is plotted by the black curve on the left-hand side of Fig. 1. More recent methods for determining C_M have been discussed and reported in Nathan's [20] and Cross' [22] researches, both of them are as functions of the speed and the spin of the ball.

Furthermore, such model has been extended for any arbitrary directions of $\boldsymbol{\omega}$ and $\mathbf{V}$ by means of

$$\mathbf{F}_\mathrm{M} = \frac{1}{2}\rho A C_\mathrm{M} \sin\phi V^2 \hat{\mathbf{u}} \tag{5}$$

where $\hat{\mathbf{u}} = \frac{\boldsymbol{\omega}\times\mathbf{V}}{|\boldsymbol{\omega}\times\mathbf{V}|}$ gives the direction of $\mathbf{F}_\mathrm{M}$, and $\sin\phi$ is introduced by considering $\boldsymbol{\omega}$-component perpendicular to $\mathbf{V}$ is the effective component of $\mathbf{F}_\mathrm{M}$, and that it varies smoothly for angles $0^\mathrm{o} \le \phi \le 90^\mathrm{o}$ between $\boldsymbol{\omega}$ and $\mathbf{V}$ [8,18].

Regarding volume forces, Coriolis and centrifugal forces are often omitted [8,15,22], however, we have estimated the final deflection that produces each force involved in the dynamics of a spinning throw. Table 1 shows forces, their expressions, the values of acceleration (α) and the final deflection (dξ) that produces them assuming the ball remains flying around a time $t = 0.5$ s such that $\mathrm{d}\xi = \alpha t^2/2$ (according to the kinematic equations of motion [10]). Acceleration values are computed in the next way: gravitational force varies because of the equatorial bulge [21]; values of centrifugal acceleration are computed by varying colatitude angle ψ; Coriolis values are obtained changing V values in throws where $\mathbf{V}\perp\mathbf{\Omega}$ so that they are the maximum values that produce such force; it is a similar case to Magnus acceleration, whose values are computed by taking $\phi = 90^\mathrm{o}$, i.e., $\boldsymbol{\omega}\perp\mathbf{V}$ and only varying values of ω, $k = \frac{1}{2}\rho A$; in turn, drag acceleration considers values of initial velocity inside the limits mentioned in Section 1.1.

[1] A reverse direction of Magnus force (anti-Magnus effect) has been reported for smooth balls like those used in soccer games [25], and smooth spheres simulating baseballs in Briggs experiments [29] and references therein.

Approximation of centrifugal–in both radial and tangential components with respect to the Earth's surface–and Coriolis forces are completed according to Taylor [10]. In them, Earth is taken like a sphere rotating at constant angular velocity $\mathbf{\Omega}$ and moving without acceleration in its translational motion, R is the radius of the Earth, ψ is the colatitude angle at which the ball is positioned, and $\hat{\mathbf{s}}$ represents the direction of the throw projected in the $x-y$ plane.

It can be seen that centrifugal force–in both radial and tangential components–could produce a final deflection larger than two millimeters (∼6% of the radius of a baseball); thus we consider this distance is enough to introduce centrifugal force in the movement equations. An example of why we limit the work to forces producing deflections of more than 1 mm can be seen when thinking on the consequences in a real pitch [15]: direction of a hit ball considering the centrifugal force will be very different from that without it[2], but it will not be so for Coriolis force.

In this way, motion of a pitched baseball is defined by

$$m\dot{\mathbf{V}} = kC_{\mathrm{d}}V^2(-\hat{\mathbf{V}}) + kC_{\mathrm{M}}\frac{V}{\omega}\boldsymbol{\omega}\times\mathbf{V} + m\mathbf{g} + m\Omega^2 R\sin^2\psi\,\hat{\mathbf{g}} + m\Omega^2 R\sin\psi\cos\psi\,\hat{\mathbf{s}}, \tag{6}$$

where drag C_{d} and Magnus C_{M} coefficients are functions of velocity $\mathbf{V}$ and angular velocity $\boldsymbol{\omega}$, respectively, like on the left of Fig. 1. In turn, angular velocity $\boldsymbol{\omega}$ of the ball does not change in magnitude ω nor in spin axis $\hat{\boldsymbol{\omega}}$. Thus, developing the cross product of the Magnus force term in (6) and defining γ as the angle between a vector points to West and direction of y axis, equations of motion into components are written as

$$\begin{aligned}
\dot{V}_x &= \frac{k}{m}V\left[\frac{C_{\mathrm{M}}}{\omega}(\omega_y V_z - \omega_z V_y) - C_{\mathrm{d}}V_x\right] + \Omega^2 R\sin\psi\cos\psi\sin\gamma \\
\dot{V}_y &= \frac{k}{m}V\left[\frac{C_{\mathrm{M}}}{\omega}(\omega_z V_x - \omega_x V_z) - C_{\mathrm{d}}V_y\right] + \Omega^2 R\sin\psi\cos\psi\cos\gamma \\
\dot{V}_z &= \frac{k}{m}V\left[\frac{C_{\mathrm{M}}}{\omega}(\omega_x V_y - \omega_y V_x) - C_{\mathrm{d}}V_z\right] + \Omega^2 R\sin^2\psi - g,
\end{aligned} \tag{7}$$

where $V = \sqrt{Vx^2 + Vy^2 + V_z^2}$, $\omega = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$, and $g \equiv |\mathbf{g}|$ is a function of ψ because of the equatorial bulge[21].

2. Calculation of the Magnus effect

2.1. Energetic analysis

Firstly, we note Eq. (6) can be rewritten as

$$\frac{d\mathbf{V}}{dt} = \alpha V\mathbf{V} + V(\boldsymbol{\beta}\times\mathbf{V}) + \mathbf{\Gamma} \tag{8}$$

where $\alpha = -\frac{kC_D}{m}$, $\boldsymbol{\beta} = \frac{kC_M}{m\omega}\boldsymbol{\omega}$, and $\mathbf{\Gamma} = \frac{\mathbf{F}_g + \mathbf{F}_{cf}}{m}$. As a comparison, Eq. (8) is the analogous of the Lorentz equation of a charged particle in electromagnetism theory [31], so that the first term (drag force) is the equivalent of the electric field, while the second one (Magnus force) is related to the magnetic field.

Developing the scalar product between Eq. (8) and $\mathbf{V}$,

$$\begin{aligned}
\frac{d\mathbf{V}}{dt}\cdot\mathbf{V} &= \left(\alpha V\mathbf{V} + V(\boldsymbol{\beta}\times\mathbf{V}) + \mathbf{\Gamma}\right)\cdot\mathbf{V} \\
&= \alpha V\mathbf{V}\cdot\mathbf{V} + V(\boldsymbol{\beta}\times\mathbf{V})\cdot\mathbf{V} + \mathbf{\Gamma}\cdot\mathbf{V}.
\end{aligned} \tag{9}$$

Then, Eq. (9) can be simplified to

$$\frac{d}{dt}\left(\frac{V^2}{2}\right) = \alpha V^3 + \mathbf{\Gamma}\cdot\mathbf{V} \tag{10}$$

by definition of scalar product and orthogonality of vector product.

Thus, Eq. (10) indicates that angular velocity–involved in $\boldsymbol{\beta}$–only changes the direction of the ball but it does not play a role in the work carried out by the system, like the action of the magnetic field on a charged particle [31].

2.2. Final deflection as a function of the angular velocity

In this way, we are interested in estimating the deflection of the ball at the end of the trajectory that is caused by Magnus force for all possible initial conditions. Eq. (7) are a coupled system of three first order nonlinear equations, for which it is complicated to obtain an analytic solution, and consequently simulating pitches and measuring the deflections. However, system (7) can be solved numerically by Runge–Kutta fourth order method (RK4) for guaranteeing the convergence

[2] It is true that magnitude of tangential centrifugal force reaches large values only at a few intervals of colatitude but it is precisely this assumption which justifies considering that force in movement equations, since baseball games are distributed at different colatitude angles.

of solutions [32]. To compute the model, drag coefficient estimation of Adair [15] was fitted by a Boltzmann function using reduced chi-squared statistic and simplex method (with OriginPro 8 program), obtaining

$$C_{\mathrm{d}}(V) = 0.29 + 0.22\left[1 + e^{(V-32.37)/5.2}\right]^{-1}, \tag{11}$$

which is plotted on the left of Fig. 1 by the blue sigmoidal curve. Root-mean-square error was under 7×10^{-3} for all values of V.

Fig. 2 shows the deflection at home produced by each component of angular velocity $\boldsymbol{\omega}$ for pitches with different initial velocities $\mathbf{V} = (0, V_y, 0)$, $V_y \in [30, 50]$ m/s, and at normal density air conditions (1.22 kg/m^3), colatitude of 90°, gravity of 9.8 m/s^2, and considering the ball has a mass of 142 g and a diameter of 7.16 cm (according to the Official Baseball Rules [19]). Deflections are calculated by means of resting final positions from those ones obtained by the solving system (7) without Magnus force terms–also via RK4. Deflections in the same direction of angular velocity components are not plotted because they are zero for all values by construction of the model [18], e.g., Magnus acceleration is zero in first equation of system (7) when angular velocity takes values distinct to zero only in x-component, such that $\boldsymbol{\omega} = (\omega_x, 0, 0)$.

Values of ω_z produce larger deflections in x-axis (dx) than those caused by ω_y in a proportion of $\mathrm{dx}_{\omega_z} \sim 10\mathrm{dx}_{\omega_y}$; this is because ω_z is related to V_y, which takes the largest values of all velocity components. A similar case occurs at z-axis where ω_x is connected to V_y. On the other hand, deflections in y-axis permit to observe the role of V_z in throws: maximum deflections produced by ω_x are slightly larger than those one by ω_z because graphics are computed with initial V_z equal to zero, therefore it reaches larger values than V_x by gravitational force. However, difference between deflections would be lower or larger for the cases when V_z takes positive or negative initial values, respectively.

Moreover, this is connected with the inverse shape of up-left and center-left meshes in Fig. 2, whose angular velocity terms are related to V_z in system (7); in turn, up-right and bottom-left graphics present the same case in relation to V_y, whereas center-right and bottom-right do it for V_x.

Thus, for common pitches with V_z close to zero, deflection in x-axis is mainly caused by ω_z and poorly by ω_y, while deflections in y and z axes are practically caused by ω_x.

3. Method for reconstructing trajectories

3.1. Characterization of a trajectory by three points

As a starting point, we propose to deal with the movement Eq. (7) omitting Magnus force terms, which is valid by Eq. (10) in Section 2.1.

The resulting system is

$$\begin{aligned}
\ddot{x} &= -\frac{kC_{\mathrm{d}}}{m} V V_x + \Omega^2 R \sin\psi \cos\psi \sin\gamma \\
\ddot{y} &= -\frac{kC_{\mathrm{d}}}{m} V V_y + \Omega^2 R \sin\psi \cos\psi \cos\gamma \\
\ddot{z} &= -\frac{kC_{\mathrm{d}}}{m} V V_z + \Omega^2 R \sin^2\psi - g,
\end{aligned} \tag{12}$$

where $\ddot{x} \equiv \dot{V}_x$, $\ddot{y} \equiv \dot{V}_y$, $\ddot{z} \equiv \dot{V}_z$ are the accelerations of the ball in x, y and z components, respectively. Although it is significantly less complicated than system (7) but it is still difficult to solve its initial value problem (simulating pitches) by traditional analytic methods[3] [34,35] and even more for its boundary value problem (reconstruction of trajectories) [35–37].

However, a theorem[4] [32,36] guarantees the uniqueness of solution for the two-point BVP of each equations of system (12). This means that a trajectory computed by model (12) can be characterized by only two positions as functions of time, namely $\boldsymbol{\xi}(t_1) = \boldsymbol{\xi}_1$ and $\boldsymbol{\xi}(t_3) = \boldsymbol{\xi}_3$, with $t_1 < t_3$. Such result should not be surprising if we analyze from the point of view of physics of a typical ballistic throw: a projectile thrown from point $\boldsymbol{\xi}_1$ at time t_1 that goes over a second point $\boldsymbol{\xi}_3$ at t_3. It is not difficult to think there is only an initial velocity (V_x, V_y, V_z) that can make the ball pass at positions (x_2, y_2, z_2), (x_3, y_3, z_3) at times t_2, t_3, respectively[5].

Now, what happens if Magnus force is introduced in system (12)? Meshes of Fig. 2 indicate that ball deflections are not oscillatory for $|\boldsymbol{\omega}|$. This motivates to think only another point $\boldsymbol{\xi}_2$, with $t_1 < t_2 < t_3$ is necessary to represent a complete trajectory. However, it is important to notice we do not intend to say the three-point BVP of system (7) has a unique solution!. Evidently it is wrong since equations of system (7) are coupled and a component of $\boldsymbol{\omega}$ can produce the same effect of another one, which can be seen in Fig. 2: final deflections remain ascending (or decreasing) for $|\boldsymbol{\omega}|$, but it is not the same case for $\boldsymbol{\omega}$ in all meshes. In spite of it, magnitude of oscillatory deflections are tiny compared with those always ascending; thus we suggest the real solution of the problem can be found.

[3] We only have found a work (Sayed, 2015 [33]) whose similar system to (12) is solved analytically.

[4] The theorem is shown by Keller [36] but can be also found in Burden and Faires [32] on page 672. Eq. (12) do not satisfy the first condition but it is not required because they do not dependent on $\boldsymbol{\xi}$ (and therefore it is the trivial case).

[5] Indeed, there could be a little interval of possible initial conditions because of drag coefficient behavior; however, it should be very little since speed of the projectile does not change a lot.

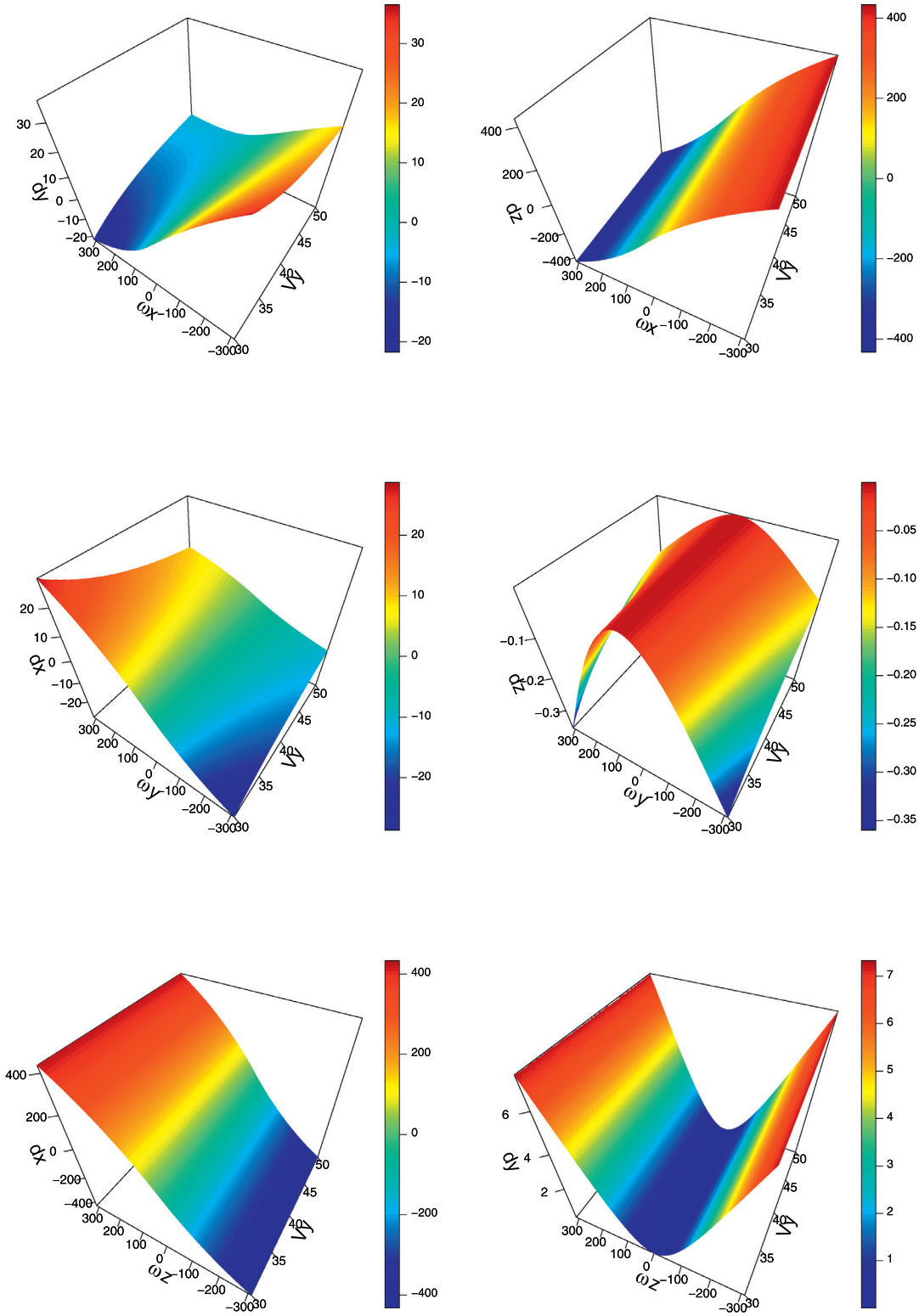

Fig. 2. Final deflection as a function of velocity and angular velocity for balls thrown with $\mathbf{V} = (0, V_y, 0)$, $V_y \in [30, 50]$ m/s. Up: Spin of $\boldsymbol{\omega} = (\omega_x, 0, 0)$. Center: Spin of $\boldsymbol{\omega} = (0, \omega_y, 0)$. Down: Spin of $\boldsymbol{\omega} = (0, 0, \omega_z)$; ω_x, ω_y, $\omega_z \in [-310, 310]$ rad/s.

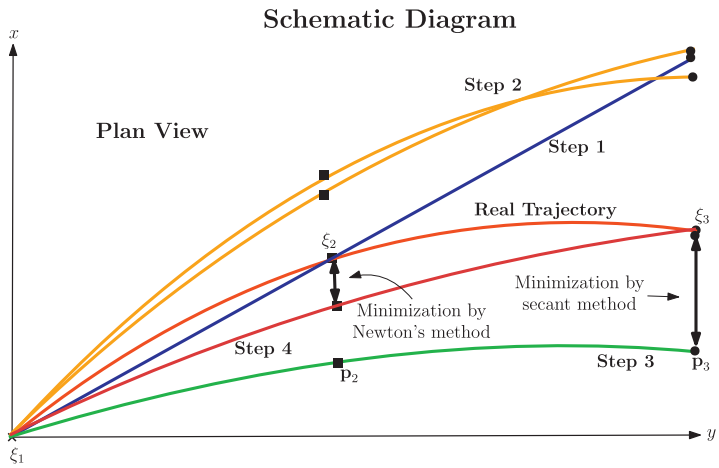

Fig. 3. Schematic diagram of the algorithm. Newton–Raphson method is responsible for minimizing the distance between the data $\mathbf{p}_2$ and the proposed point $\boldsymbol{\xi}_2$ by approximating $\mathbf{V}$, while secant method does it for points at time t_3 by approximating $\boldsymbol{\omega}$.

3.2. The algorithm

Based on this proposal, initial conditions of any trajectory can be obtained by minimizing the function

$$f(\mathbf{V}, \boldsymbol{\omega}) = \sum_{i=1}^{3} ||\boldsymbol{\xi}_i - \mathbf{p}_i||, \tag{13}$$

where $\mathbf{p}_i \equiv \mathbf{p}_i(t_i)$ are the ball positions at time t_i of a possible trajectory, and $||\cdot||$ refers to the euclidean distance. Moreover, position of the ball at point $\boldsymbol{\xi}_1$ is known, since is the origin of the trajectory; thus Eq. (13) could be simplified to

$$f(\mathbf{V}, \boldsymbol{\omega}) = \sum_{i=2}^{3} ||\boldsymbol{\xi}_i - \mathbf{p}_i||. \tag{14}$$

Now the question becomes how to minimize it. To answer this, we can observe that a more accurate estimation of the velocity can be made at the beginning of a throw because largest deflections originated by Magnus force occur at the end of the trajectory (and then dynamics of the ball at points closest to the initial point is practically determined by its initial velocity). Thus, minimization of distance $||\boldsymbol{\xi}_2 - \mathbf{p}_2||$ in function (14) shall be related to finding values of initial velocity, whereas the angular velocity shall be connected to $||\boldsymbol{\xi}_3 - \mathbf{p}_3||$.

Such observation together with the common use of the shooting method to solve the two-point BVP of equations similar to (12)–using Newton–Raphson method to obtain a solution for the initial value problem (IVP) [32]–led to construct a method composed of two interrelated parts or with two objectives, as described in the schematic diagram of Fig. 3.

The flow diagram of the algorithm is shown in Fig. 4. It consists of five steps:

- **Step 0**. Data points and parameters of the method are defined.
- **Step 1**. Shooting method is applied to obtain the first velocity approximation $\mathbf{V}^0$, using points $\boldsymbol{\xi}_1$ and $\boldsymbol{\xi}_2$, considering the assumptions mentioned in this Section; additionally, such assumptions permit us using Newton-Raphson method to solve the IVP.
- **Step 2**. Initial values for angular velocity are defined. We suggest $\boldsymbol{\omega}^0 = (-50, -50, -50)$ rad/s and $\boldsymbol{\omega}^1 = (50, 50, 50)$ rad/s (waiting for having initial approximations covering the search space but inside the range of possible solutions).
- **Step 3**. The wide range of angular velocity solutions demands solving the IVP by a slower method in convergence to Newton–Raphson; this leads to use the secant method, where $\boldsymbol{\omega}^k-$ components are approximated according to the results of Section 2.2, and with the restriction $|\boldsymbol{\omega}^k| < 310$ rad/s for the solutions to remain inside the search space.
- **Step 4**. New velocity $\mathbf{V}^k$ values are obtained in a similar way to Step 2 but considering the obtained $\boldsymbol{\omega}^k$ values.
- **Step 5**. Steps 3 and 4 are repeated as in an iterative manner until minimizing function (14) below the tolerance (Tol) defined in Step 0.

4. Results

4.1. Synthetic trajectories

The method is applied to some synthetic trajectories, which are generated by solving (7) as in Section 2.2 and with the same parameters. Tests were carried out by using R-Statistics program in a Lenovo 400 RAM, 250 GB, Intel-Core i5 computer.

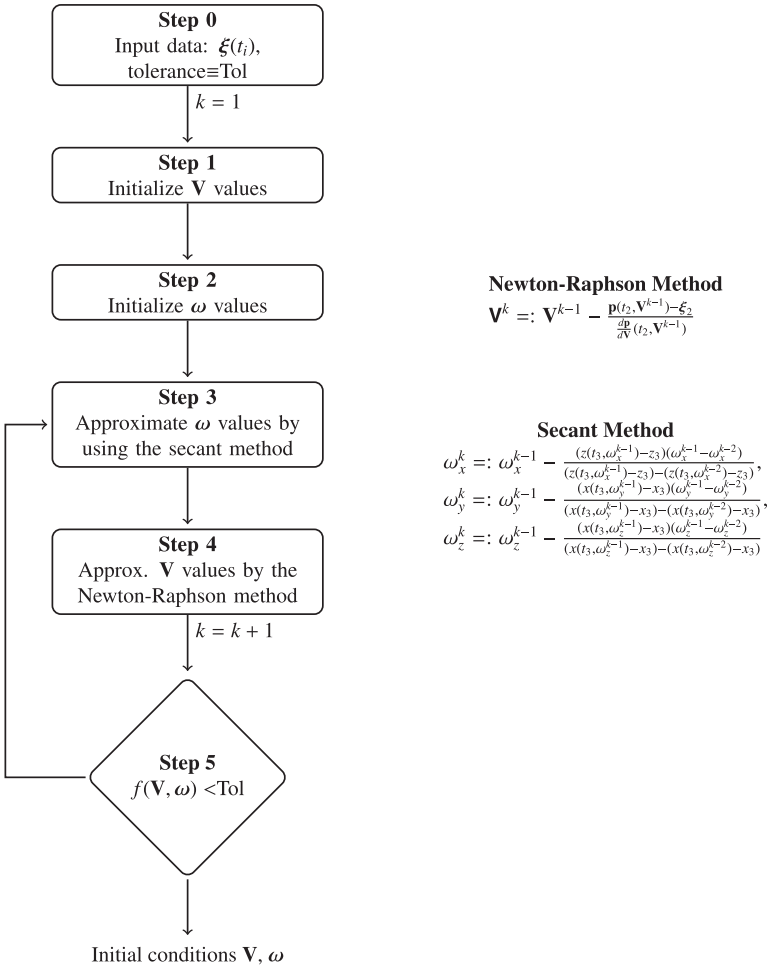

Fig. 4. Flow diagram of the algorithm. Steps are represented by the schematic diagram of Fig. 3.

Table 2
First test. Four trajectories are obtained by varying the location of the ball position $\boldsymbol{\xi}_2$, e.g., up-black results are those obtained by locating y_2 at three-quarter parts of distance between y_3 and y_1. Initial velocity $\mathbf{V}$ and angular velocity $\boldsymbol{\omega}$ are obtained by minimizing the function $f(\mathbf{V}, \boldsymbol{\omega})$ in a k number of iterations.

Parameters	Results			
ξ_2 location	$\mathbf{V}$ (m/s)	ω (rad/s)	$f(\mathbf{V}, \boldsymbol{\omega})$	k iter.
$\frac{3}{4}(y_3 - y_1)$	$(-1.8 \times 10^{-3}, 45.00, 1.2 \times 10^{-3})$	$(300.8, 39.9, -2.8)$	8.4×10^{-5}	27
$\frac{1}{2}(y_3 - y_1)$	$(-1.2 \times 10^{-3}, 45.00, 6.1 \times 10^{-4})$	$(301.0, 39.8, -2.7)$	6.4×10^{-5}	13
$\frac{1}{3}(y_3 - y_1)$	$(-7.9 \times 10^{-4}, 45.00, 5.7 \times 10^{-4})$	$(301.1, 39.8, -2.6)$	5.1×10^{-5}	9
$\frac{1}{4}(y_3 - y_1)$	$(-5.9 \times 10^{-4}, 45.00, 3.3 \times 10^{-4})$	$(301.1, 39.7, -2.5)$	2.5×10^{-5}	8
Initial Values:	(0, 45, 0)	(300, 0, 0)		

Under these conditions the computational time was ~ 3 s for solutions obtained around $k = 10$, whereas ~ 6 s when $k = 25$; see Tables 2 and 3.

Table 2 shows the results for the reconstruction of a ball, which is thrown at time $t_0 = 0$ from the origin $\boldsymbol{\xi}_1 = (0, 0, 0)$ and reaches home at position $\boldsymbol{\xi}_3 = (2.6 \times 10^{-3}, 17.05, -3.3 \times 10^{-1})$–measuring in meters–at time $t_3 = 0.388$, and varying data $\boldsymbol{\xi}_2$. The stop rule is by minimizing the function (14) below the tolerance Tol$= 10^{-4}$ m. The pitch corresponds to the initial conditions $\mathbf{V} = (0, 45, 0)$ m/s, $\boldsymbol{\omega} = (300, 0, 0)$ rad/s. To complement it, the distance between the obtained and real trajectories throughout the flight of the ball is plotted in Fig. 5.

As it can be seen, better solutions for such study are reached when using y_2 approximates to y_1. This is because of the construction of the method: choosing a value of $\boldsymbol{\xi}_2$ close to the initial point $\boldsymbol{\xi}_1$ allows to obtain a better approximation of $\mathbf{V}$ in Step 1 of Algorithm 4; in turn, a good first approximation of the ball velocity is very important because it allows to dedicate all iterations to approximate $\boldsymbol{\omega}$ and only improving $\mathbf{V}$ values. In this way, better (regarding function f) and faster

Table 3
Results for the second test. Same notation as in Table 2.

Parameters	Results			
ξ_2 location	$\mathbf{V}$ (m/s)	ω (rad/s)	$f(\mathbf{V}, \boldsymbol{\omega})$	k iter.
$\frac{3}{4}(y_3 - y_1)$	(2.002, 39.00, 0.999)	(28.6, 49.9, 195.6)	8.8×10^{-5}	26
$\frac{1}{2}(y_3 - y_1)$	(2.001, 39.00, 0.999)	(28.9, 58.6, 195.7)	9.3×10^{-5}	13
$\frac{1}{3}(y_3 - y_1)$	(2.000, 39.00, 0.999)	(29.2, 64.2, 196.0)	9.1×10^{-5}	10
$\frac{1}{4}(y_3 - y_1)$	(2.000, 39.00, 0.999)	(29.8, 67.0, 196.1)	9.8×10^{-5}	8
Initial values:	(2, 39, 1)	(30, 80, 200)		

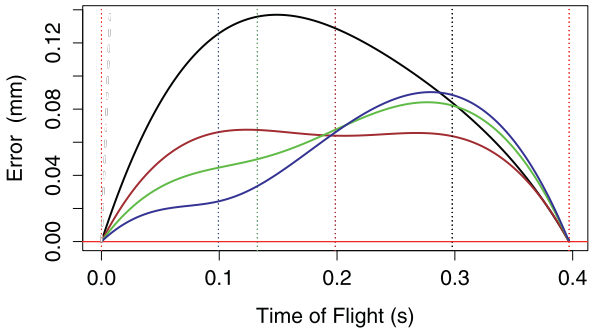

Fig. 5. First test. Error between trajectories along the pitch. Vertical dashed lines tagged t_1 and t_3 with red, and times when the ball crosses the midpoint $\boldsymbol{\xi}_2$ of the trajectory (according to Table 2), namely, 1/4 parts in blue, 1/3 parts in green, 1/2 parts in brown, and 3/4 parts in black. Error trajectories lines are obtained by using the midpoint $\boldsymbol{\xi}_2$ tagged with the corresponding color. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

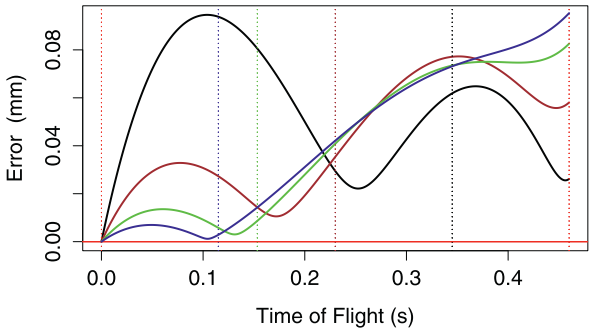

Fig. 6. Second test. Error between trajectories along the pitch. Same notation as in Fig. 5.

(low number of iterations) solutions are obtained from the methodology of Newton–Raphson ($\mathbf{V}$) and secant methods ($\boldsymbol{\omega}$) [32,38].

However, for all positions of $\boldsymbol{\xi}_2$, the obtained trajectories are always close to the real one (error$<$ 0.2 mm), although the error presents oscillations. This behavior is originated because the method found the values that minimize only the distances at the intermediate point (tagged with dashed-color lines) and at the end point (red-dashed line).

Second test consists in a throw with initial conditions $\mathbf{V} = (2, 39, 1)$ m/s, $\boldsymbol{\omega} = (30, 80, 200)$ rad/s, which produce the ball cross home-plate at $\boldsymbol{\xi}_3 = (-3.1 \times 10^{-1}, 17.08, -9.6 \times 10^{-1})$, with $t_3 = 0.461$, taking $\boldsymbol{\xi}_1 = (0, 0, 0)$ as the initial position. It shows the other side of choosing a y_2 point very close to y_1: this could produce a slight divergence of the obtained trajectory at the end of it, while trajectories with y_2 points near the midpoint remain in convergence; see left-hand side of Fig. 6.

According to Table 3, the method seems to converge to better solutions around the same iterations than for the first case of study. More precisely, in both cases values of $\mathbf{V}$ are approximated with an accuracy of 10^{-3} m/s for all $\boldsymbol{\xi}_2$ points chosen.

About $\boldsymbol{\omega}$ approximations, although the method does not bring solutions very close to data, it approximates ω_x and ω_z sufficiently to fit the real trajectory; in turn, ω_y values are very far from the optimum but this is not reflected because of the role of ω_y in the meshes of Fig. 2.

With the aim of having a comparison algorithm, we have carried out the first test substituting Steps 3 and 4 in Fig. 4 by the Levenberg-Marquardt nonlinear least-squares fitting algorithm [39]. Such algorithm has been used in problems of reconstruction of baseball trajectories but in a very different way, obtaining good approximations with computational time below

Table 4
Results for the first test of Section 4.1 by applying the Levenberg–Marquardt nonlinear least-squares fitting algorithm.

Parameters	Results			
ξ_2 location	$\mathbf{V}$ (m/s)	ω (rad/s)	$f(\mathbf{V}, \boldsymbol{\omega})$	k iter.
$\frac{3}{4}(y_3 - y_1)$	$(-1.4 \times 10^{-3}, 45.00, -1.2 \times 10^{-5})$	$(301.0, 35.9, -0.8)$	6.7×10^{-5}	16
$\frac{1}{2}(y_3 - y_1)$	$(-9.0 \times 10^{-4}, 45.00, -7.0 \times 10^{-6})$	$(301.1, 36.1, -0.7)$	3.3×10^{-5}	14
$\frac{1}{3}(y_3 - y_1)$	$(-5.9 \times 10^{-4}, 45.00, -4.3 \times 10^{-6})$	$(301.1, 36.1, -0.6)$	1.1×10^{-5}	14
$\frac{1}{4}(y_3 - y_1)$	$(-4.4 \times 10^{-4}, 45.00, -1.0 \times 10^{-6})$	$(301.1, 36.2, -0.6)$	2.6×10^{-7}	14
Initial values:	(0, 45, 0)	(300, 0, 0)		

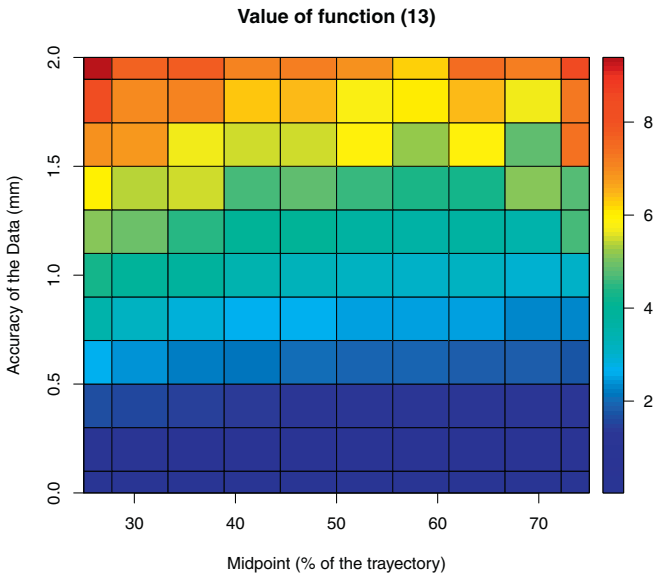

Fig. 7. Colormap of the error between points of the original and proposed trajectories expressed in terms of the function (13) for different values of both midpoints $\boldsymbol{\xi}_2$ and accuracy of the original data. Midpoints are expressed in percent of the total time of flight, while accuracy is expressed by the maximum absolute value it can take in each direction.

1 s [20,23]. The implementation was completed by means of the *lsqcurvefit*() function in R statistics program, requiring *pracma* library, using the default parameters and the initial approximations $\mathbf{V} = (0, 40, 0)$, $\boldsymbol{\omega} = (50, 50, 50)$.

Table 4 shows the results for this test. In general terms, fitness and number of iterations are similar for both the Levenberg–Marquardt algorithm and the algorithm proposed in Section 3.2 but in contrast with the last one, the Levenberg–Marquardt algorithm does not seem to be sensitive to the choice of $\boldsymbol{\xi}_2$. Computational time was around 10 s when $k = 14$, and 12 s when $k = 16$, which indicates that both algorithms require the same order of CPU time for obtaining good solutions. All of this suggests that the Algorithm 4 could be improved for obtaining better approximations in lower CPU times.

4.2. Algorithm's robustness

Noise was introduced to the synthetic data in order to check the robustness of the Algorithm 4, thinking in an application of it. A colormap of the distance between points of the original and proposed trajectories is shown in Fig. 7. In contrast with tests of Section 4.1, function (13) must be chosen because of the noise in all $\boldsymbol{\xi}_i$ data. The grid was made by taking 10 midpoints $\boldsymbol{\xi}_2$ equally spaced between 1/4 and 3/4 of the trajectory (x−axis), and adding uniform random values inside $[-2, 2]$ mm to the original data (y−axis), which correspond to relative errors of 0–9.7% with respect to the radius of the ball. In this way, the disturbed data have an accuracy of $\pm$ 2 mm for each direction of motion, and the noise in a determined point is independent from every other point; such noise characteristics are obtained by the systems used for tracking baseballs in Major League Baseball [20,23].

In Fig. 7, each point of the grid represents the average of 200 runs of the algorithm. The algorithm is robust when increasing the noise in the original data, namely, when the distance between the points of the original and proposed trajectories is below 10 mm for all accuracy of the data (even when it reaches the 2 mm), and seems to have a linear behavior with respect to it. However, sensitivity to the midpoint $\boldsymbol{\xi}_2$ can be seen at choosing values closer to the beginning and end of the trajectory. In fact, it seems to have a quadratic behavior with origin at $\boldsymbol{\xi}_2$=1/2. These are acceptable approximations in comparison with those obtained by the existing techniques; concretely, root-mean-squares deviation values in trajectories obtained by Nathan [20] are in the order of 10 mm.

5. Conclusions

A method to reconstruct trajectories of spinning baseballs pitches was designed. We found that only three ball positions define a trajectory. This result is applied in an algorithm based on the shooting method, which obtains the initial conditions of synthetic trajectories by minimizing the distance between three data points of the original trajectory and those of the proposed trajectory, with high accuracy in low times. Robustness of the algorithm is shown when uniform random noise is introduced to the data. Specifically, the sum of errors in the objective function increases linearly with respect to the added noise, obtaining acceptable results in comparison with existing techniques. In addition, the Levenberg–Marquardt algorithm was used in substitution of the shooting method obtaining slightly better results in the same order of computational time for synthetic trajectories. All of these results leads to consider that our method provides good predictions for comparison with future experiments of throws with high spins.

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